SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Artificial Intelligence

Practical 3

Aim: Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

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CODE A:

T074 Ai Pract 3.py - C:/Users/PRIYANKA/Downloads/T074 Ai Pract 3.py (3.13.1)

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```python
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Importing the dataset
dataset = pd.read_csv('iris.csv')
X = dataset.iloc[:, [0, 3]].values
y = dataset.iloc[:, -1].values

# Splitting the dataset into the Training set and Test set
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_state = 0)

# Feature Scaling
#since our dataset containing character variables
#we have to encode it using LabelEncoder
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)

# Predicting the Test set results
y_pred = classifier.predict(X_test)
print("Predicted Test Results : ", y_pred)
print("~"*20)

# Making the Confusion Matrix
from sklearn.metrics import confusion_matrix, accuracy_score
ac = accuracy_score(y_test, y_pred)
print("Model Accuracy : ", ac*100, "%")
print("~"*20)
cm = confusion_matrix(y_test, y_pred)
print("Model Confusion Matrix : ")
print(cm)
```

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OUTPUT:

IDLE Shell 3.13.1

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```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
============ RESTART: C:/Users/PRIYANKA/Downloads/T074 Ai Pract 3.py ===========
Predicted Test Results :  ['Iris-virginica' 'Iris-versicolor' 'Iris-setosa' 'Iris-virginica'
 'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-versicolor'
 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor'
 'Iris-versicolor' 'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa'
 'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
 'Iris-virginica' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
 'Iris-virginica' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
 'Iris-versicolor' 'Iris-setosa']
~~~~~~~~~~~~~~~~~~~~
Model Accuracy :  100.0 %
~~~~~~~~~~~~~~~~~~~~
Model Confusion Matrix :
[[11  0  0]
 [ 0 13  0]
 [ 0  0  6]]
>>>
```

Ln: 20 Col: 0

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SUBJECT NAME: Artificial Intelligence

CODE B:

T074 Ai Pract 3B.py - C:/Users/PRIYANKA/Downloads/T074 Ai Pract 3B.py (3.13.1)

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve, auc
import matplotlib.pyplot as plt

print("Libraries imported")

# Load Dataset
pima_df = pd.read_csv("diabetes.csv")
print("Dataset Loaded")

# Predictor and Target Variables
X = pima_df.drop("Outcome", axis=1)
y = pima_df["Outcome"]

# Standardize the Features
std = StandardScaler()
X = std.fit_transform(X)
print("Transformed Data")
print(X)

# Split the Dataset
test_size = 0.30
seed = 7

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=test_size,
    random_state=seed
)

# Create Gaussian Naive Bayes Model
model = GaussianNB()

# Train the Model
model.fit(X_train, y_train)
print("\nModel Trained Successfully")
```

T074 Ai Pract 3B.py - C:/Users/PRIYANKA/Downloads/T074 Ai Pract 3B.py (3.13.1)

```python
# Make Predictions
predicted = model.predict(X_test)
print("\nPredicted Values:")
print(predicted)

# Confusion Matrix
cm = confusion_matrix(y_test, predicted)
print("\nConfusion Matrix:")
print(cm)

# Accuracy
accuracy = accuracy_score(y_test, predicted)
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Prediction Probabilities
y_predictProb = model.predict_proba(X_test)
print("\nPrediction Probabilities:")
print(y_predictProb)

# ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_predictProb[:, 1])
roc_auc = auc(fpr, tpr)
print("\nROC AUC Score:", roc_auc)

# Plot ROC Curve
plt.figure(figsize=(8, 6))
plt.plot(
    fpr,
    tpr,
    color="darkorange",
    label="ROC Curve (AUC = %0.2f)" % roc_auc
)
plt.plot([0, 1], [0, 1], color="navy", linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Receiver Operating Characteristic (ROC)")
plt.legend(loc="lower right")
plt.grid(True)
plt.show()
```

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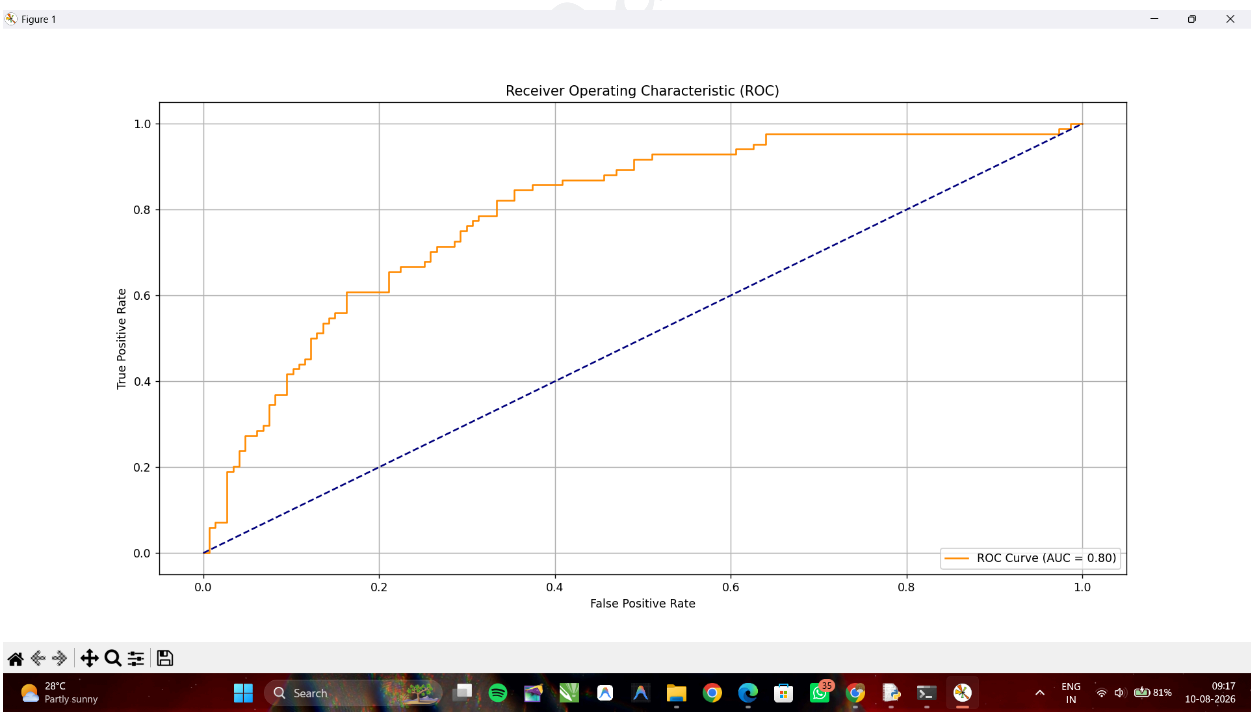
OUTPUT:

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=========== RESTART: C:/Users/PRIYANKA/Downloads/T074 Ai Pract 3B.py ===========
Libraries imported
Dataset Loaded
Transformed Data
[[ 0.63994726  0.84832379  0.14964075 ...  0.20401277  0.46849198
   1.4259954 ]
 [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078
  -0.19067191]
 [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732
  -0.10558415]
 ...
 [ 0.3429808   0.00330087  0.14964075 ... -0.73518964 -0.68519336
  -0.27575966]
 [-0.84488505  0.1597866  -0.47073225 ... -0.24020459 -0.37110101
   1.17073215]
 [-0.84488505 -0.8730192   0.04624525 ... -0.20212881 -0.47378505
  -0.87137393]]

Model Trained Successfully
GaussianNB()

Predicted Values:
[0 1 1 0 0 0 0 0 1 0 1 0 1 1 1 0 0 0 0 0 1 0 1 1 0 0 0 1 1 0 0 0 0 1 0 0 0
 0 1 1 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 0 0 0 0 1 0 1 1 0 0 1 1 0 1 1 0 0
 1 0 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 1 1 1 0 0 1 0 0 1 0 0 1 0 0 1 0 0 1 1 0
 1 0 0 1 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0 0
 1 1 0 1 0 0 0 0 1 0 0 1 1 0 0 0 1 0 0 0 0 1 0 1 1 0 0 0 1 1 1 0 0 0 0 0 0
 0 0 0 0 1 1 1 1 1 0 0 1 0 0 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 0 0 1 1 0 0 0 1
 1 0 0 1 0 0 1 0 0]

Confusion Matrix:
[[116  31]
 [ 29  55]]

Accuracy: 74.03%

Prediction Probabilities:
Squeezed text (231 lines).

ROC AUC Score: 0.7973760932944606
```



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